

Sub-Sec	Subject	Qs	Marks
c-1	Elements of Mechanical Engg.	8	8
c-2	Elements of Production Engg./Mfg.	6	6
c-3	Automobile Engineering	6	6
TOTAL		20	20

c-1 ELEMENTS OF MECHANICAL ENGINEERING (8 Questions | 8 Marks)

■ Applied Mechanics

- Friction: laws — $F = \mu N$; $\mu_s > \mu_k$; angle of friction $\phi = \arctan(\mu)$; self-locking condition
- Centroid: for standard shapes (rectangle, triangle, circle, semicircle)
- Simple Machines: $MA = \text{Load/Effort}$; $VR = \text{effort distance/load distance}$; $\eta = MA/VR$
- Screw Jack: $VR = 2\pi/p$ ($l = \text{lever length}$, $p = \text{pitch}$); Wheel & axle: $VR = R/r$
- Systems of Pulleys: single fixed, single movable, compound; Projectile motion equations
- Work, Power, Energy: $W = F \cdot d$; $P = W/t$; $KE = \frac{1}{2}mv^2$; $PE = mgh$

■ Strength of Materials

- Stress $\sigma = F/A$; Strain $\epsilon = \Delta L/L$; Young's modulus $E = \sigma/\epsilon$ (Hooke's law)
- Stress-strain curve: proportional limit, elastic limit, yield point, UTS, fracture
- Temperature stress: $\sigma = E\alpha\Delta T$; Composite sections: $\sigma_{\text{steel}}/\sigma_{\text{concrete}} = E_{\text{steel}}/E_{\text{concrete}}$ (same strain)
- Elastic constants: $E = 2G(1+\nu) = 3K(1-2\nu)$; $G = \text{shear modulus}$; $K = \text{bulk modulus}$; $\nu = \text{Poisson's ratio}$
- Principal stresses: $\sigma_{1,2} = (\sigma_x + \sigma_y)/2 \pm \sqrt{[(\sigma_x - \sigma_y)^2/4 + \tau^2]}$; max shear $= \frac{1}{2}(\sigma_1 - \sigma_2)$
- SFD & BMD: simply supported ($R_A + R_B = \text{total load}$); cantilever (fixed end max moment)
- Resilience: strain energy per unit volume $U = \sigma^2/2E$; Proof resilience, modulus of resilience
- Columns & Struts: Euler's formula $P_{cr} = \pi^2 EI / (L_e)^2$; slenderness ratio

■ Thermodynamics

- Zeroth Law: thermal equilibrium basis of temperature measurement
- First Law: $Q = \Delta U + W$; $dU = mC_v dT$; for closed system $\Delta U = Q - W$
- Second Law: Kelvin-Planck & Clausius statements; entropy always increases
- Steam: dry/wet/superheated; dryness fraction x ; properties from steam tables
- Cycles: Otto (petrol) $\eta = 1 - 1/r^\gamma$; Diesel $\eta = 1 - (1/r^\gamma)(\gamma - 1) \times [(rc^\gamma - 1)/(\gamma - 1)]$
- Thermodynamic processes: isothermal ($T = \text{const}$), adiabatic ($Q = 0$), isobaric ($P = \text{const}$), isochoric ($V = \text{const}$)

■ Heat Transfer

- Conduction: $Q/t = kA(T_{\text{hot}} - T_{\text{cold}})/L$; Fourier's law; thermal conductivity k
- Convection: $Q = hA(T_s - T_\infty)$; Newton's law of cooling; natural vs forced
- Radiation: $Q = \epsilon\sigma AT^4$; Stefan-Boltzmann law; emissivity ϵ (blackbody $\epsilon = 1$)

■ Fluid Mechanics

- Properties: density ρ , specific gravity, viscosity μ (dynamic), $\nu = \mu/\rho$ (kinematic)
- Newtonian fluids: $\tau = \mu(du/dy)$; Non-Newtonian: pseudoplastic, dilatant, Bingham plastic
- Bernoulli: $P/\rho g + v^2/2g + z = \text{constant}$ (along streamline, ideal fluid)
- Flow types: Laminar ($Re < 2000$), Turbulent ($Re > 4000$); $Re = \rho v D / \mu$
- Continuity: $A_1 v_1 = A_2 v_2$; Flow measurement: Pitot tube, Venturimeter
- Darcy-Weisbach: $h_f = f L v^2 / (2gD)$; f from Moody chart

c-2 ELEMENTS OF PRODUCTION ENGINEERING (6 Questions | 6 Marks)

■ Workshop / Machine Tools

- Lathe: turning, facing, taper turning, threading; $MRR = \pi D N \times f \times d$; Cutting speed $v = \pi D N / 1000$
- Drilling: twist drill; operations — drilling, boring, reaming, countersinking, tapping
- Milling: up milling (conventional) vs down milling (climb); types of cutters
- Shaper: single point tool, reciprocating; Planer: for larger workpieces

■ Measuring Instruments

- Vernier Caliper: least count = 1 MSD – 1 VSD (typically 0.02mm or 0.05mm)
- Micrometer: $LC = \text{pitch} / \text{no. of divisions on thimble} = 0.01\text{mm}$
- Sine Bar: $\sin \theta = H/L$; used with slip gauges; Plug, snap, ring gauges — Go/No-Go

■ Joining Processes

- Welding types: arc (SMAW, MIG, TIG), gas (oxy-acetylene), resistance (spot, seam)
- Soldering: below 450°C; Brazing: above 450°C — filler above copper-zinc alloy

■ Material Science

- Mechanical properties: strength, hardness, toughness, ductility, malleability, fatigue, creep
- Hardness testing: Brinell (BHN), Vickers (VHN), Rockwell (HRA/B/C); impression measurement
- Heat Treatment: Annealing (soften), Normalizing (refine grain), Hardening (quench), Tempering (reduce brittleness), Carburizing (surface hardening)

■ Engineering Drawing

- First angle projection (ISO): object between viewer and plane; used in India/Europe
- Third angle projection (ANSI): plane between viewer and object; used in USA
- Orthographic views: Front, Top, Side; Isometric: 30° axes, true length along axes

■ Foundry & Plastics

- Patterns: solid, split, loose piece, follow board, match plate, gated; pattern allowances
- Moulding materials: green sand, dry sand, CO₂ sand; core making
- Plastics: thermoplastic (remoldable) vs thermosetting (permanent); molding: injection, compression, blow, extrusion

■ Industrial Management

- Org. structures: line, functional, line-and-staff, matrix
- Leadership qualities: decision-making, communication, motivation, technical competence

- Quality control: SQC, control charts (X-bar, R-chart), acceptance sampling
- Productivity = Output/Input; methods improvement: time & motion study

c-3 AUTOMOBILE ENGINEERING (6 Questions | 6 Marks)

- Power Flow: Engine → Clutch → Gearbox → Propeller shaft → Differential → Half shafts → Wheels
- Clutch types: single plate, multi-plate, cone; purpose: smooth engagement/disengagement of power
- Gearbox: sliding mesh, constant mesh, synchromesh; gear ratios; overdrive
- Differential: allows wheels to rotate at different speeds during cornering; bevel gear arrangement
- Brakes: drum brakes (wheel cylinder, shoes), disc brakes (caliper, pad); hydraulic, mechanical, air brakes
- Cooling system: air cooling (fins) vs water cooling (radiator, thermostat, water pump, coolant)
- Lubrication: splash, pressure, dry sump systems; SAE grades; functions of lubricant
- Wheels & Tyres: bias ply vs radial; tyre markings (e.g. 185/65 R15); rim types
- Steering: Ackermann principle; power steering; steering angles: camber, caster, toe-in, king pin
- E-Vehicles: BEV (battery only), PHEV, HEV (hybrid); regenerative braking; major EV companies
- Major auto industries: Maruti Suzuki, Tata Motors, Mahindra, Hero MotoCorp, Bajaj (India); globally Toyota, VW, GM